Jaiveer Bassi

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Experience

Physical Intelligence May 2026 – Present
Robot Operator *San Francisco, CA*

- Generate high-quality demonstration data for general-purpose robotics foundation models by teleoperating robotic arms through diverse manipulation, sorting, and assembly tasks with precision and consistency.
- Perform quality control on AI-driven robot performance by evaluating model outputs against benchmarks, identifying failure modes and edge cases, and annotating task execution data to meet training-quality thresholds for model development.
- Triage and diagnose recurring hardware-software faults across a distributed robotics stack, tracing failures through service logs to isolate root cause and documenting findings for engineering teams.

Handshake AI March 2026 – June 2026
Contracted AI Response Evaluator *Remote*

- Evaluated LLM-generated code and technical responses across AI/ML and software engineering domains, scoring on correctness, instruction following, and writing quality.

Kaiser Permanente Regional Laboratory Oct 2023 – June 2024
Software Developer *Berkeley, CA*

- Developed full-stack web application using React, Express, RESTful APIs, and SQL for uploading, processing, and viewing clinical lab instrument data across multiple departments.
- Triaged and resolved bug reports related to faulty instrument data recording, diagnosing root causes and documenting failure patterns to improve data reliability across the pipeline.

Projects

Article: Simulating Real Robot Failures on a Breadboard – *CAN Bus Diagnostics Rig | C++, Python, SocketCAN, Embedded Systems*

- Built a 3-node CAN bus on custom hardware (Arduino/MCP2515 nodes, USB-CAN interface) and induced real bus-off, arbitration, and connector faults, diagnosing each via SocketCAN error-state analysis with a reproducible fault catalog.
- Code, wiring notes, and full fault catalog on GitHub.

Article: A $30 Teleoperation Rig – *GELLO | C++, Python, Arduino, Servo Control*

- Rebuilt the core mechanism behind GELLO-style robot teleoperation for under $30, substituting potentiometer-based leader encoding for the servo-as-encoder approach used in production rigs, preserving direct joint-space control with no inverse kinematics.
- Code, wiring notes, and build guide on GitHub.

UR5e Fault Injection & Diagnostics Harness – *Real-Time Robot Control Interface | Python, RTDE, Docker, URSim*

- Built a control and fault-injection harness against UR5e's production RTDE interface using UR's official simulator, reproducing a real segmentation fault from a production traceback and characterizing which connection failures are catchable in application code versus requiring external process supervision.
- Logged synchronized joint telemetry at fixed sample rate with automated gap detection to recover diagnostic evidence from failures that crash the process before any exception is raised.

Imitation Learning Failure Mode Analysis – *Robot Learning Research | Python, Imitation Learning, Closed-Loop Policy Evaluation*

- Built the codebase for published research quantifying how demonstration failure modes degrade closed-loop policy performance, directly informing how bad demonstrations propagate through imitation learning pipelines.

Robot Fleet Diagnostics & Root Cause Analysis Tool – *Fault Classification System | Python, Streamlit, ML, Anomaly Detection*

- Built a synthetic robot fleet log analyzer performing fault classification, recurrence detection, anomaly detection, and rule-based root cause analysis with a real-time Streamlit fleet health dashboard.

Robot Grasp Failure Annotation Pipeline – *CV Data-Ops System | Python, Labelbox SDK, COCO JSON*

- Built a data-ops pipeline for annotating robot arm grasp events with bounding boxes, keypoints, and failure mode classification labels, automated via Labelbox Python SDK with COCO JSON export.

Robot Telemetry Pipeline (Kubernetes) | *Python, Kubernetes, Redis Streams, PostgreSQL*

- Built a multi-stage robot telemetry pipeline on Kubernetes using Redis Streams and PostgreSQL, with idempotent upserts guaranteeing exactly-once processing under at-least-once delivery.
- Designed a fault-injection harness that kills pods and partitions the network across 8 chaos scenarios, measuring data loss and recovery time from the data itself.

Research Publications

Trajectory-Smoothness Curation Metrics Under Episode-Length Controls | DOI: 10.13140/RG.2.2.26129.60000

- Audited published trajectory-smoothness curation metrics against episode-length controls on robomimic..

Demonstration Failure Modes in Closed-Loop Policy Performance | DOI: pending

- Quantified how different types of bad demonstrations degrade imitation learning policy performance in closed-loop evaluation.

Brain Tumor Classification with Pretrained CNNs in PyTorch | DOI: 10.13140/RG.2.2.21638.28484

- Classified brain tumors into four categories from 7,000+ MRI images using ResNet-18 transfer learning.

Efficient Packaging and Deployment of AI Models for Edge Inference | DOI: 10.13140/RG.2.2.23010.59844

- Achieved 4x model size reduction and 50-70% faster inference via quantization, pruning, and lightweight containerization.

Transferability of Adversarial Attacks Across ML Models | DOI: 10.13140/RG.2.2.16410.76489

- Demonstrated 99%+ cross-architecture attack success rates evaluating FGSM, PGD, and CW attacks on CIFAR-10.

Technical Skills

Robotics & Diagnostics: Teleoperation, VR-based robot control, imitation learning, closed-loop policy evaluation, fault triage, root cause analysis, log parsing, telemetry analysis
Languages: Python, TypeScript, JavaScript, Swift, SQL, C++, Bash
AI/ML: PyTorch, OpenAI API, LangChain, HuggingFace, Labelbox, Computer Vision, RAG Pipelines
Software: React, Node.js, Flask, Docker, Git, Linux, RESTful APIs, AWS

Education

Grand Canyon University May 2024 – Oct 2025
MS in Computer Software Engineering *Phoenix, AZ*

University of Silicon Valley May 2021 – Aug 2023
BS in Computer Science *San Jose, CA*

Certifications

HackerRank Software Engineer Certificate (Mar 2026)
Cisco Network Support and Security (Sep 2025)
IBM Artificial Intelligence (Aug 2025)
Stanford AI/ML in Healthcare (Jun 2025)